

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15A.2

OPTICAL ACTIVITY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15A.2: Optical Activity
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15A.2

Q1. 1. (a) Describe how a polarimeter is used to distinguish between two optical isomers.

[3 Marks]

WCH14/01/Jun23/Q5(a)

Q2. (b) Explain why a racemic mixture (racemate) exhibits no optical activity.

[2 Marks]

WCH14/01/Jun23/Q5(b)

Q3. 2. MCQ: A solution containing 0.10 mol dm^{-3} of an enantiomer rotates plane-polarised light by $+12^\circ$. What rotation is observed for a 50:50 mixture of both enantiomers?

[1 Marks]

WCH14/01/Jan22/Q2

- A. $+12^\circ$
- B. -12°
- C. $+6^\circ$
- D. 0°

SECTION B: WORKED MARK SCHEME — 15A.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Describe how a polarimeter is used to distinguish between two optical isomers. 1. Plane-polarised light is passed through a solution of each enantiomer. 2. One enantiomer rotates the plane of polarisation clockwise (dextrorotatory). 3. The other enantiomer rotates the plane by an equal angle anticlockwise (laevorotatory).	[3]
Q2	Q2: (b) Explain why a racemic mixture (racemate) exhibits no optical activity. 1. A racemic mixture contains equal amounts (50:50) of both enantiomers. 2. The clockwise rotation by one enantiomer is exactly cancelled by the equal anticlockwise rotation of the other.	[2]
Q3	Q3: 2. MCQ: A solution containing 0.10 mol dm^{-3} of an enantiomer rotates plane-polarised light by $+12^\circ$. What rotation is observed for a 50:50 mixture of both enantiomers? D (0° due to cancellation in racemic mixture).	[1]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15A.2**FAQ #1: Why does a racemate show 0° optical rotation?****Category: Polarimetry****COMMON EXAMINER TRAP / MISTAKE:**

Stating that the molecules in a racemate do not rotate light at all.

MODEL ANSWER / EXAMINER EXPECTATION:

Each individual enantiomer molecule rotates light, but because equal numbers of (+) and (-) enantiomers are present, the opposite rotations cancel out macroscopically.